

Take-home assignment 3 – Dynamic Linear Models with R.

PART I: Filtering and smoothing for the random walk plus noise model.

Solutions of questions 1-2-3 and 5 are due by April 14, 2026

For these exercises, **see LAB 3, posted on Blackboard!**

Install and load package **dlm**

Exercise.

Consider again the Nile data (measurements of the annual flow of the river Nile at Ashwan 1871-1970), available in R (`> ?Nile`).

First, plot the data. The series clearly appears non-stationary, presenting a quite evident change point. A *local level* model, i.e. a random walk plus noise, may be used to capture the main change point *and* other minor changes in the level of the Nile river. Let us consider the following random walk plus noise model

$$\begin{aligned}y_t &= \theta_t + v_t, & v_t &\sim \mathcal{N}(0, \sigma_v^2) \\ \theta_t &= \theta_{t-1} + w_t, & w_t &\sim \mathcal{N}(0, \sigma_w^2)\end{aligned}$$

with the due assumptions. To start with, assume that the variances are known, $\sigma_v^2 = 15100$, $\sigma_w^2 = 1470$. (In fact, they are estimated by MLE; see the last question below). As the initial distribution, let $\theta_0 \sim \mathcal{N}(1000, 1000)$.

1. FILTERING.

Compute the filtering states estimates $m_t = E(\theta_t | y_{1:t})$, for $t = 1, 2, \dots, T$.

Compute the corresponding standard deviations

$$\sqrt{C_t} = V(\theta_t | y_{1:t})^{1/2}$$

and plot them. Comment briefly (does $C_t \rightarrow 0$? why?).

Finally, plot the data together with the filtering state estimates and their 0.95 credible intervals.

2. ONLINE FORECASTING.

Compute the one-step ahead forecasts $f_t = E(Y_t | y_{1:t-1})$, $t = 1, \dots, T$.

Plot the data, together the one-step-ahead forecasts and their 0.95 credible intervals.

3. What is the effect of the **signal-to-noise ratio** (i.e. the ratio W/V) on the forecasts?

Repeat the exercise with different choices of σ_v^2 (observation variance) and σ_w^2 (evolution variance) and comment briefly.

4. MODEL CHECKING.

Finally, check the model, using the properties of the forecast errors.

5. SMOOTHING.

So far, for computations, we pretended that the data arrived sequentially. Now consider $(y_1, \dots, y_T)$, and provide and plot the smoothing estimate of the Nile level θ_t at time $t = 28$ together with its 95% credible interval. Then, compare them with the filtering estimates at $t = 28$.

6. MAXIMUM LIKELIHOOD ESTIMATES.

In the exercises above, we assumed that the variances σ_v^2 and σ_w^2 are known. In fact, they are unknown and have to be estimated. You may use the function **StructTS** to estimate them (see LAB 3 on BBoard). Can you provide their standard errors?

*Note: Later on, for general DLMs, we will use the package **dlm**.*